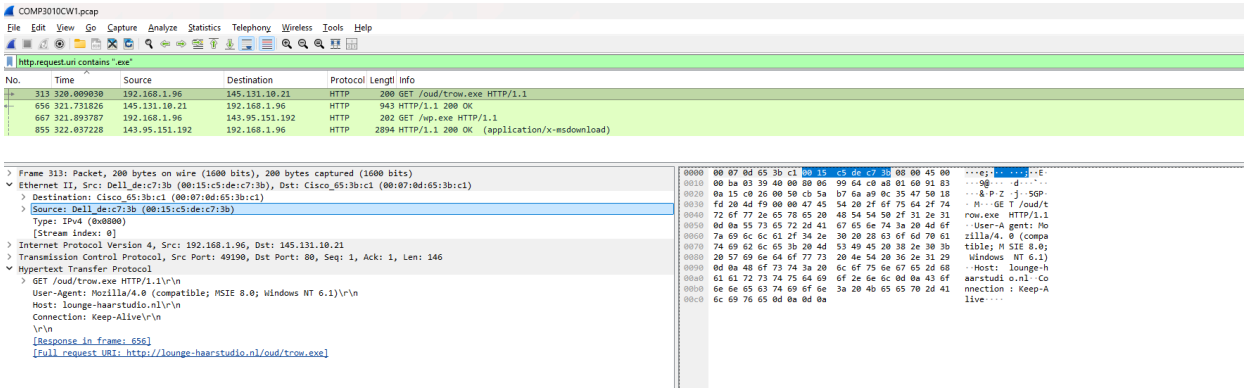
Coursework prerequisite - Validation Questions

Coursework 1 Investigation Validation Questions (ROUGH WORKING FOR EVIDENCE)

Question 1

What is the MAC address of the infected Windows computer? = “**00:15:c5:de:c7:3b**”



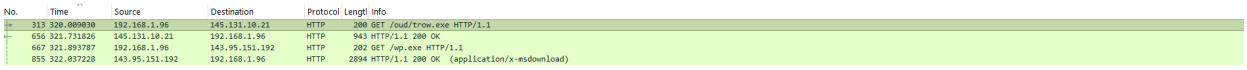
From GET of /oud/trow.exe

Frame 313: Packet, 200 bytes on wire (1600 bits), 200 bytes captured (1600 bits)
Ethernet II, Src: Dell_de:c7:3b (00:15:c5:de:c7:3b), Dst: Cisco_65:3b:c1 (00:07:0d:65:3b:c1)
    Destination: Cisco_65:3b:c1 (00:07:0d:65:3b:c1)
    Source: Dell_de:c7:3b (**00:15:c5:de:c7:3b**)
    Type: IPv4 (0x0800)
    [Stream index: 0]
Internet Protocol Version 4, Src: 192.168.1.96, Dst: 145.131.10.21
Transmission Control Protocol, Src Port: 49190, Dst Port: 80, Seq: 1, Ack: 1, Len: 146
Hypertext Transfer Protocol

The item in bold (the source getting the download) has MAC Address **00:15:c5:de:c7:3b**

Question 2

What is the IP address of the infected Windows computer? = “**192.168.1.96**”

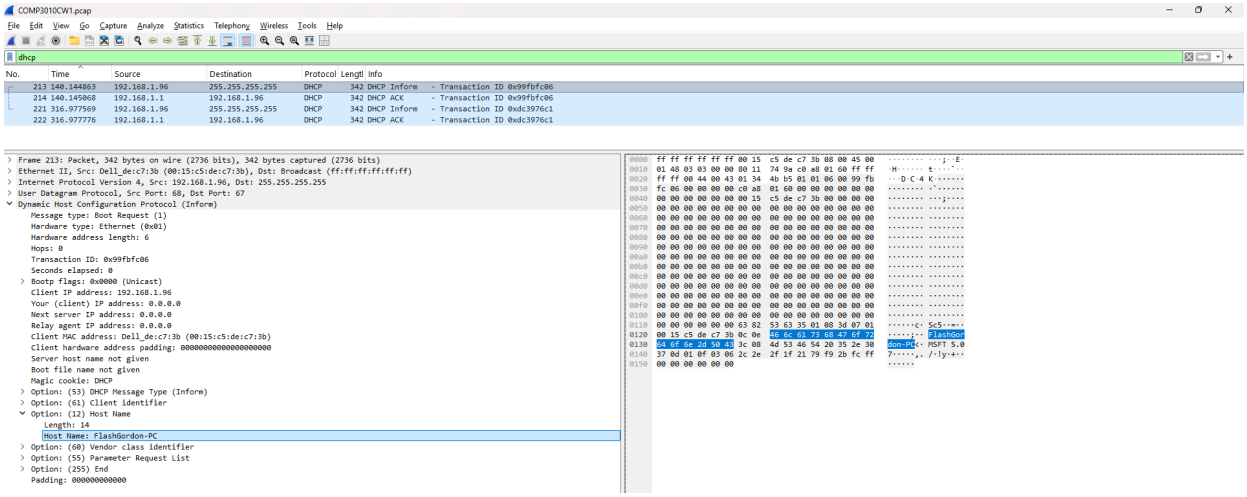


A regular IP address which made suspicious downloads and executions was source 192.168.1.96 (shown above).

Question 3

What is the hostname of the infected Windows computer? = “**FlashGordon-PC**”

Protocol hierarchy showed Dynamic Host Configuration Protocol, meaning I could filter for “dhcp”. Options here had “Host name FlashGordon-PC“.



Question 4

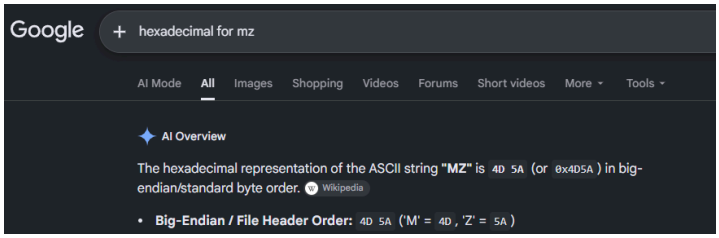
What is the date of infection? (yyyy-mm-dd) = “**2017-06-27**"

Two suspicious GET requests (313 - GET /oud/trow.exe HTTP/1.1 & GET /wp.exe HTTP/1.1) were shown on the infected IP source: 192.168.1.96 - appearing to be windows executables.

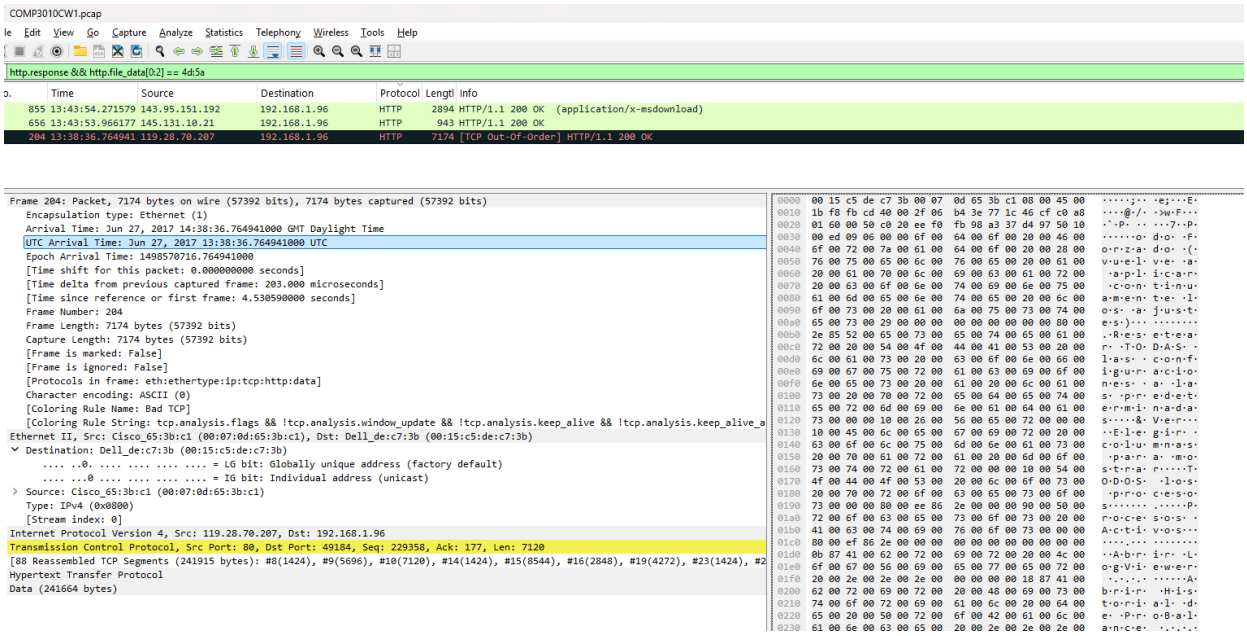
Image of the following HTTP stream…



To check when the executable content was transferred, I needed to find an HTTP call of something beginning with mz (the start of a windows executable signature). - and we know an executable was installed. The hexadecimal file signature 4D 5A for mz.



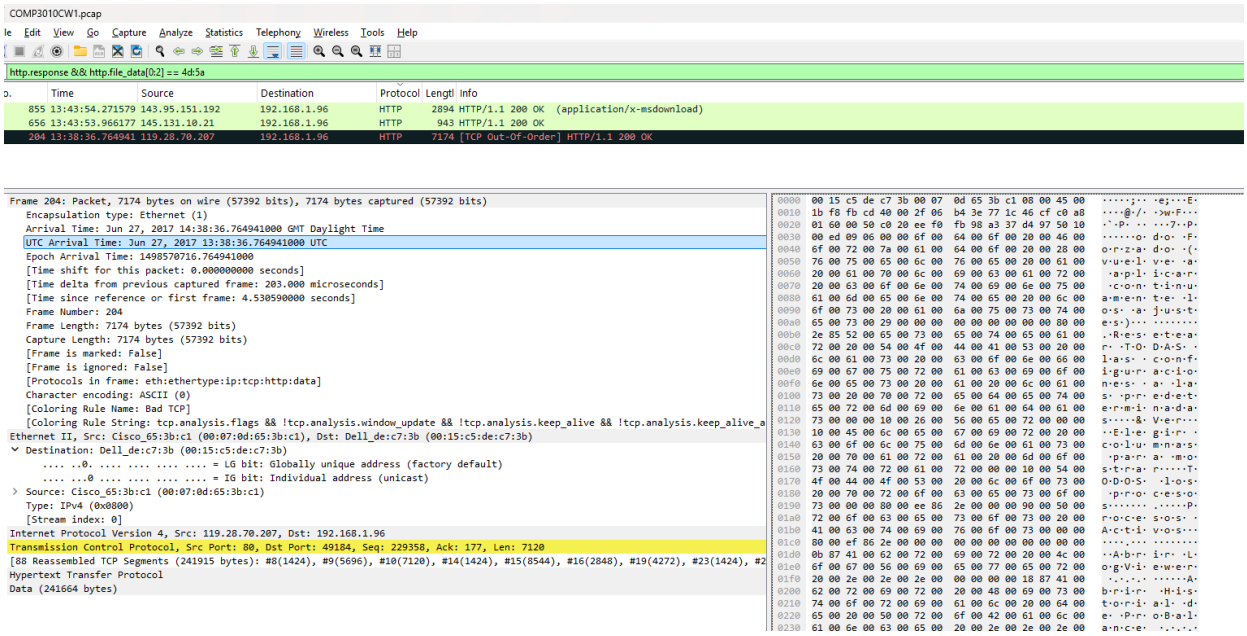
After searching for http responses starting with that hex, I found the following capture of the delivery of gerv.gunat UTC **2017-06-27T13:38:36.764941000Z** - meaning the date was 2017-06-27.



Question 5

What is the time (in UTC) of the infection? (hh:mm) = "**13:38**"

As shown above, the execution occurred at UTC 2017-06-27T13:38:36.764941000Z - meaning the time was 13:38.



Question 6

Based on the Snort and Suricata alerts, what was the computer infected with? It's a _____ variant. Enter only the keyword in uppercase = "**PUSHDO**"

After compromise, it has a Trojan, of the Pushdo variant…

```
[**] [1:23832:4] INDICATOR-OBFUSCATION non-alphanumeric javascript detected [**]
[Classification: Attempted User Privilege Gain] [Priority: 1]
06/27-13:44:08.765489 104.27.139.76:80 -> 192.168.1.96:49292
TCP TTL:53 TOS:0x0 ID:15927 IpLen:20 DgmLen:5129 DF
***AP*** Seq: 0xCECA6E07  Ack: 0x234866D1  Win: 0x1F  TcpLen: 20
[Xref => http://patriciopalladino.com/blog/2012/08/09/non-alphanumeric-javascript.html]

[**] [1:23832:4] INDICATOR-OBFUSCATION non-alphanumeric javascript detected [**]
[Classification: Attempted User Privilege Gain] [Priority: 1]
06/27-13:44:08.816660 104.27.139.76:80 -> 192.168.1.96:49294
TCP TTL:53 TOS:0x0 ID:37546 IpLen:20 DgmLen:4782 DF
***AP*** Seq: 0xAC4969A0  Ack: 0x35DA69A4  Win: 0x1F  TcpLen: 20
[Xref => http://patriciopalladino.com/blog/2012/08/09/non-alphanumeric-javascript.html]

[**] [1:15168:13] INDICATOR-COMPROMISE Suspicious .ru dns query [**]
[Classification: A Network Trojan was detected] [Priority: 1]
06/27-13:44:11.994287 192.168.1.96:51688 -> 192.168.1.1:53
UDP TTL:128 TOS:0x0 ID:3326 IpLen:20 DgmLen:54
Len: 26

[**] [1:15168:13] INDICATOR-COMPROMISE Suspicious .ru dns query [**]
[Classification: A Network Trojan was detected] [Priority: 1]
06/27-13:44:12.002972 192.168.1.96:49996 -> 192.168.1.1:53
UDP TTL:128 TOS:0x0 ID:3338 IpLen:20 DgmLen:54
Len: 26

[**] [1:29891:7] MALWARE-CNC Win.Trojan.Pushdo variant outbound connection [**]
[Classification: A Network Trojan was detected] [Priority: 1]
06/27-13:44:11.994886 192.168.1.96:49322 -> 104.27.158.125:80
TCP TTL:53 TOS:0x0 ID:23760 IpLen:20 DgmLen:1994 DF
***A**** Seq: 0xCBEB269  Ack: 0x3960D09  Win: 0x8800  TcpLen: 20

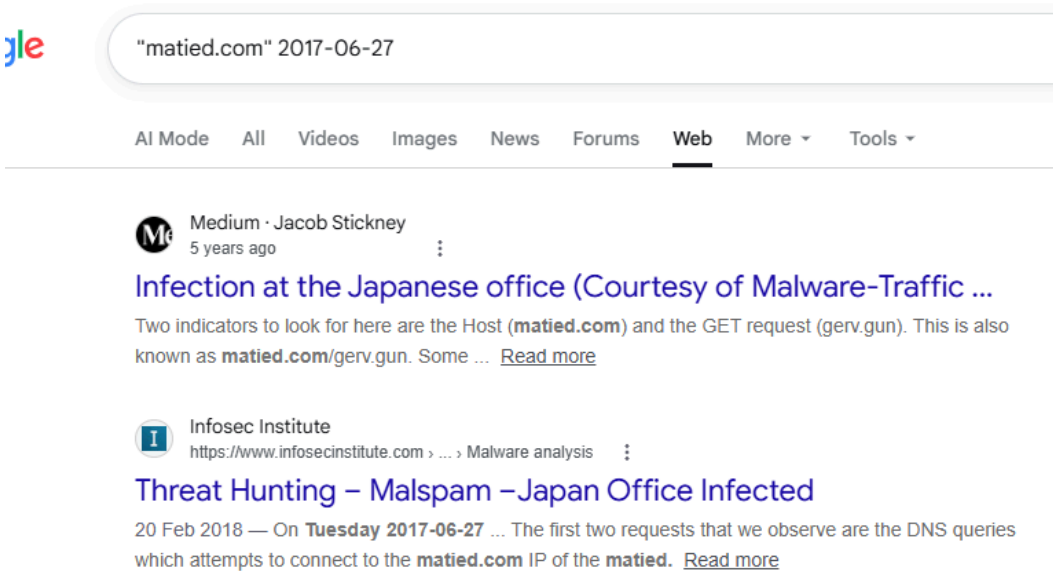
[**] [1:29891:7] MALWARE-CNC Win.Trojan.Pushdo variant outbound connection [**]
[Classification: A Network Trojan was detected] [Priority: 1]
06/27-13:44:11.994487 192.168.1.96:49318 -> 104.20.221.29:80
TCP TTL:51 TOS:0x8 ID:44202 IpLen:20 DgmLen:1289 DF
***A**** Seq: 0xEC586A97  Ack: 0x717262C7  Win: 0x8000  TcpLen: 20

[**] [1:15168:13] INDICATOR-COMPROMISE Suspicious .ru dns query [**]
[Classification: A Network Trojan was detected] [Priority: 1]
06/27-13:44:12.048047 192.168.1.96:61354 -> 192.168.1.1:53
UDP TTL:128 TOS:0x0 ID:3419 IpLen:20 DgmLen:54
```

Question 7

What is the hash of the js file that probably infected the computer? = "**64ae13592c5d8b289be81ebb50e054cef49b0bdf50ea92dd44ed611dd274150f**" OR "**59727ea201d94d471bdbb353b1281cca3c5bebf5305b140a3c017e52a37be702**"

- I searched for instances of the malware on the date I identified to find a few pages. This showed a Medium article (on already completed work) and an "infosecinstitute" page with an analysis of the malware in a real world incident.



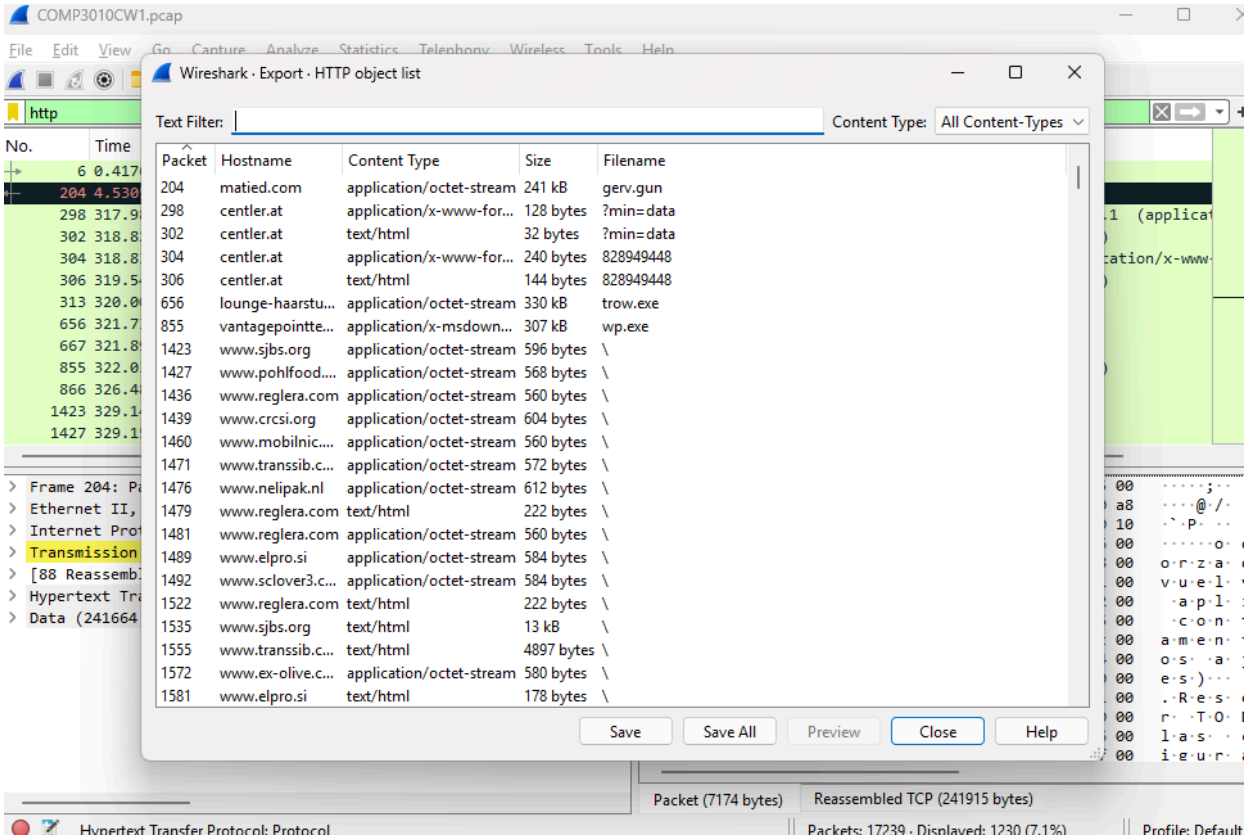
- The infosec page included some already found information on the machine, as well as malware variants and its impact. It also contained a number of references - most to general websites used (virustotal, hybrid-analysis etc.),one to an article on future protections against malware and a reference to a page on how malware can use js files from within zips.
    - https://www.infosecinstitute.com/resources/malware-analysis/threat-hunting-malspam-japan-office-infected/
- The above “infosecinstitute” page reference link took me to the “myonlinesecurity” site below. This mentioned “2017.6支払依頼書.zip Extracts to DOC0978043-2017.6.doc.js Current Virus total detections: Payload Security shows downloads from”... - though where a VirusTotal link would be there is nothing. I inspected the source of the site, though no hyperlink was available anywhere.
    - https://myonlinesecurity.co.uk/japanese-language-invoice-malspam-using-js-files-inside-zips-today/
- Prompted by the mention of a link with no hyperlink available, I checked the page’s history in the wayback machine (https://web.archive.org/) - going back to the 2017 archived version of the page. This showed the page in 2017 with the VirusTotal links showing (Link to archived version here)...
    - https://web.archive.org/web/20170909073653/https://myonlinesecurity.co.uk/japanese-language-invoice-malspam-using-js-files-inside-zips-today/
- This archived version of the ‘myonlinesecurity’ article references the following two VirusTotal files…
    - https://www.virustotal.com/gui/file/64ae13592c5d8b289be81ebb50e054cef49b0bdf50ea92dd44ed611dd274150f/detection/f-64ae13592c5d8b289be81ebb50e054cef49b0bdf50ea92dd44ed611dd274150f-1498539514
    - https://www.virustotal.com/gui/file/59727ea201d94d471bdbb353b1281cca3c5bebf5305b140a3c017e52a37be702/detection/f-59727ea201d94d471bdbb353b1281cca3c5bebf5305b140a3c017e52a37be702-1498539649

- These both contain hashes of malicious JS files
    - I would assume that the tool of researching this is useful if the executable & js file cannot be found themselves.

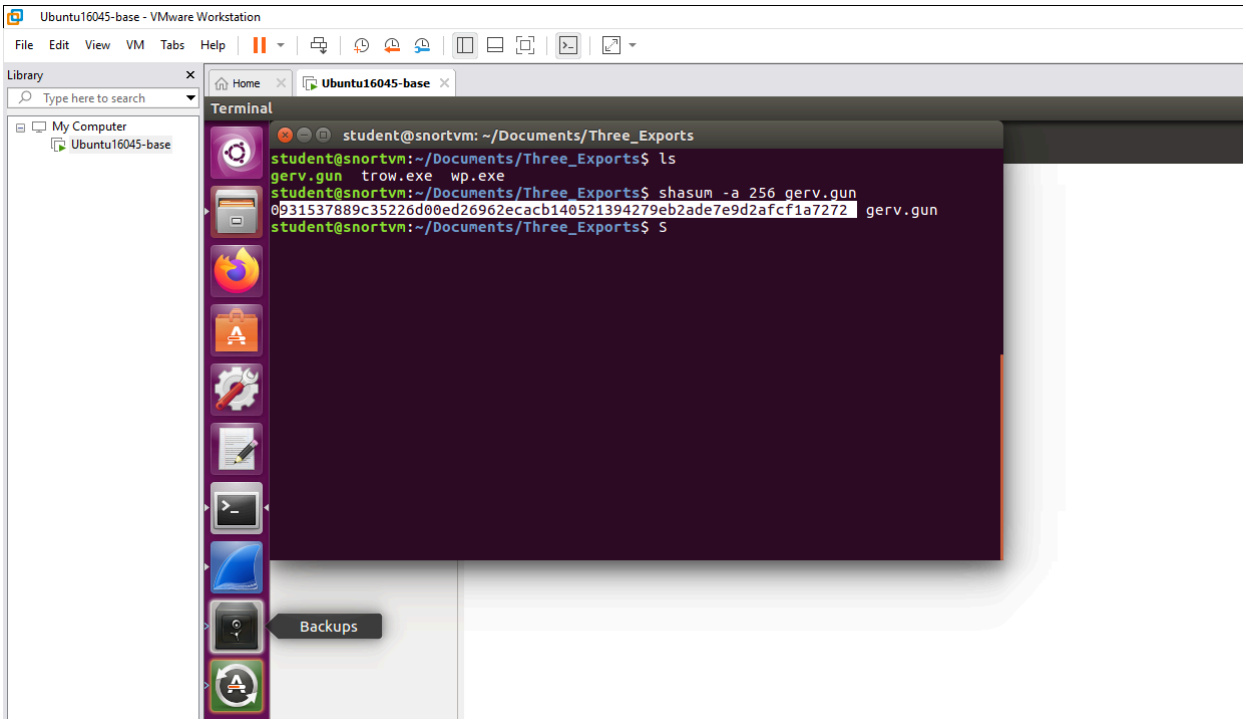
Question 8

The pcap contains 3 Windows executable files sent over HTTP. Export them from the pcap. What are the SHA256 file hashes of those 3 files? The answer for this question should be for the ****.g** file =
“**0931537889c35226d00ed26962ecacb140521394279eb2ade7e9d2afcf1a7272** “

Exported the three previously identified dodgy executables as HTTP objects…

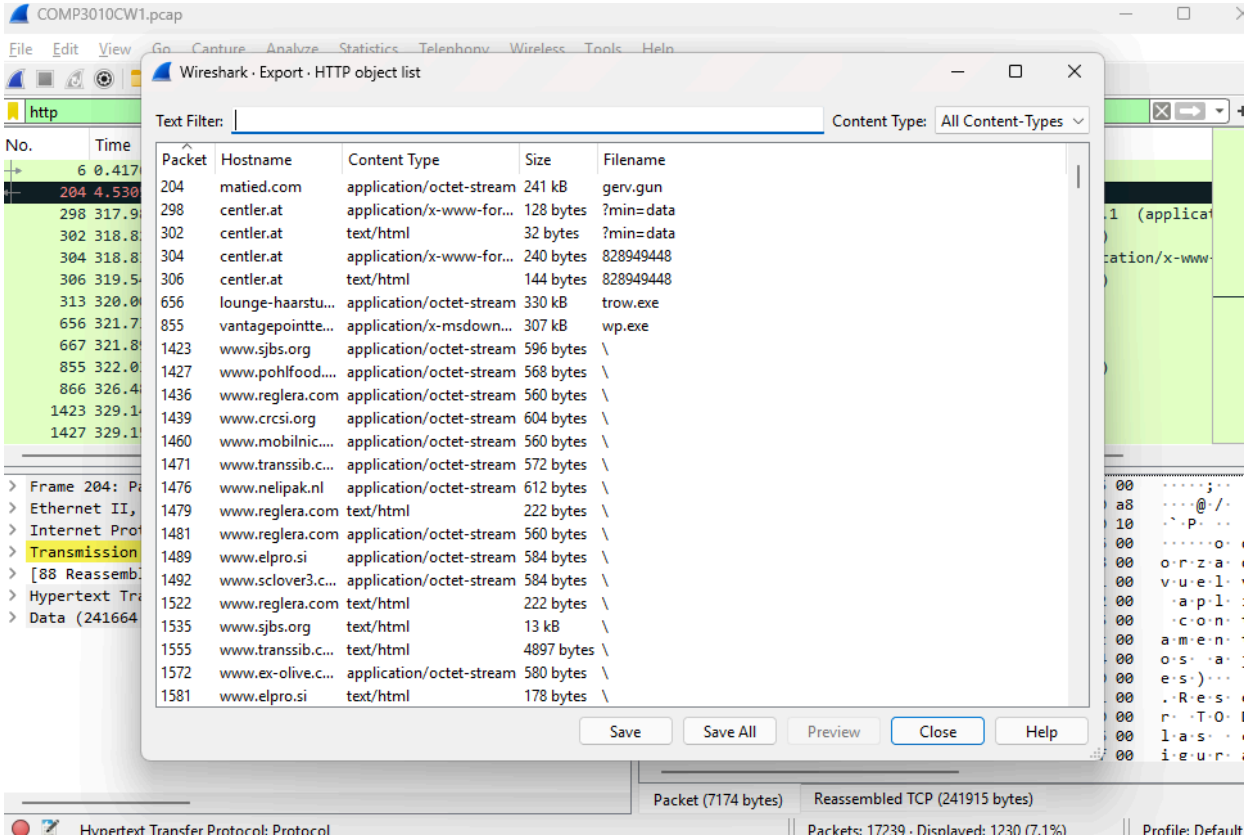


Then I ran shasum with -a (algorithm) 256 (for SHA-256 as the question requested), getting the following hash.

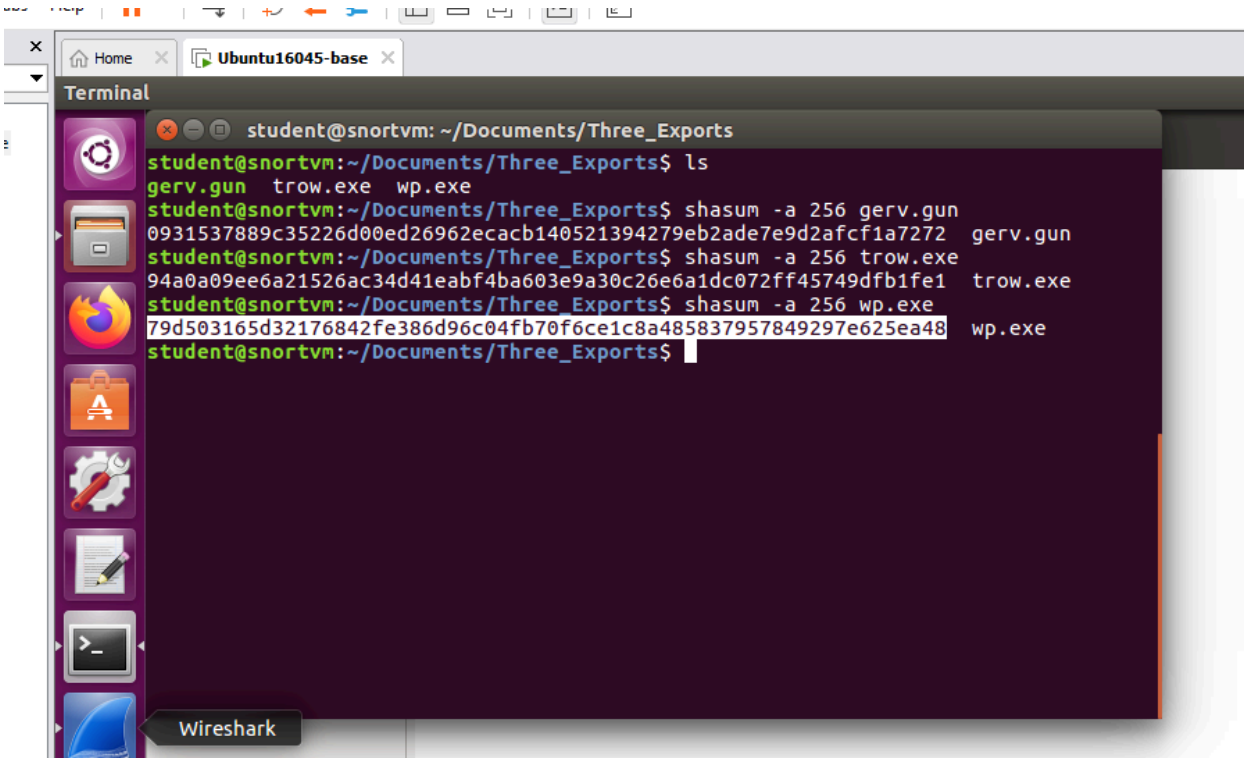


Question 9

<u>The pcap contains 3 Windows executable files sent over HTTP. Export them from the pcap. What are the SHA256 file hashes of the those 3 files? The answer for this question should be for the **.e** file</u> =
"**79d503165d32176842fe386d96c04fb70f6ce1c8a485837957849297e625ea48**"



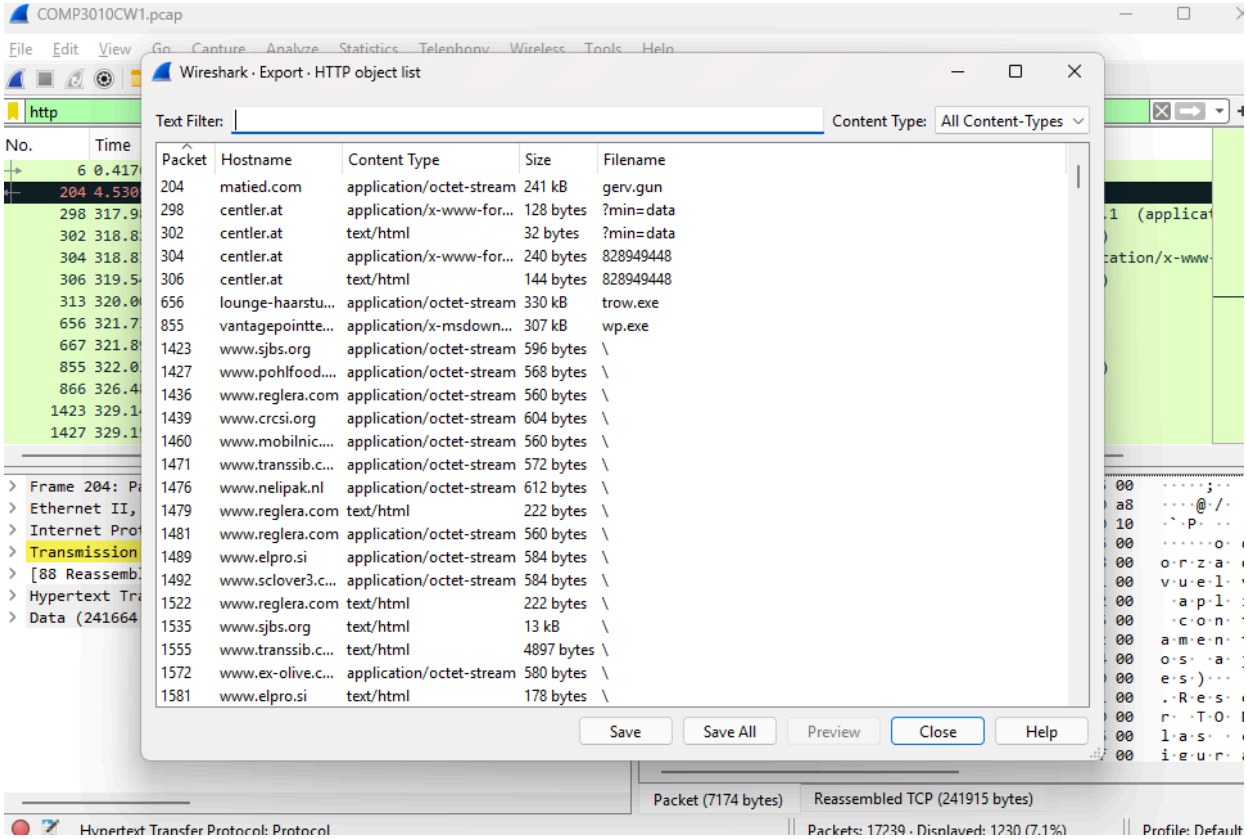
Then I ran shasum with -a (algorithm) 256 (for SHA-256 as the question requested), getting the following hash…



Question 10

The pcap contains 3 Windows executable files sent over HTTP. Export them from the pcap. What are the SHA256 file hashes of the those 3 files? The answer for this question should be for the ****.e** file =
"**94a0a09ee6a21526ac34d41eabf4ba603e9a30c26e6a1dc072ff45749dfb1fe1**"

Exported the three previously identified dodgy executables as HTTP objects…



Then I ran shasum with -a (algorithm) 256 (for SHA-256 as the question requested), getting the following hash.

